Nalvi Duro — Curriculum Vitae

Contact

Nalvi Duro, Ph.D. — Computational Biologist · Tampa, Florida

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Professional Summary

Computational biologist with a Ph.D. in cell and molecular biology and a decade of work spanning computational biophysics, bioinformatics, and large-scale genomic analysis. Leads R&D teams that build de novo long-read genome assemblies, ancestry-specific pangenomes, and reference panels, and that integrate metabolomics and PheWAS data into reproducible HPC pipelines. Author of peer-reviewed work in *Communications Biology*, *Structure*, *Proteins*, *Langmuir*, and the *Journal of Mass Spectrometry*, with a record of presenting at international meetings and coaching junior scientists.

Education

Ph.D., Cell and Molecular Biology — University of South Florida, Tampa, FL · 2020

Computational Biophysics Laboratory (advisor: Sameer Varma), Department of Cell Biology, Microbiology and Molecular Biology, College of Arts and Sciences. Dissertation: *Allosteric Control of Proteins: New Methods and Mechanisms*. Used molecular dynamics simulations to study allosteric regulation in paramyxovirus attachment proteins and protein–DNA, protein–protein and protein–lipid interactions. Conceived and implemented a method combining support vector machines, graph theory, affinity propagation and molecular dynamics to extract time-resolved allosteric signaling pathways in proteins.

Project Management Professional (PMP) — PMI · 2025

Professional Experience

Project Lead | Bioinformatician — Booz Allen Hamilton, Tampa, FL · October 2021 – Present

- Lead a bioinformatics research team: gather requirements, define scope, set milestones, manage resources and delegate tasks.
- Build de novo genome assemblies from PacBio HiFi long reads; run read-level and assembly-level QC.
- Construct ancestry-specific pangenomes and reference panels from QC-passing assemblies using HPRC pangenome methods.
- Contributor to VA Million Veteran Program genomics: a HiFi-assembly pangenome used to call variants in medically relevant genes (ACMG 2026) and a multi-ancestry GWAS of Fuchs corneal dystrophy (*Communications Biology*, 2024).
- Variant calls for novel large structural variants and missing heritability not captured by current reference genomes.

- Own data ingestion, quality assurance, filtering, normalization, imputation and integration for multi-omic datasets, including metabolomics.
- Implement and troubleshoot custom and public WDL and Snakemake pipelines on HPC clusters.
- Report findings to the scientific community through publications and conference presentations.

Chief of Molecular Modeling | Founding Member — Biological Information OS Inc., Tampa, FL · June 2020 – Present

- Design and build a digital education platform that delivers STEM material in an engaging, investigative format.
- Create 3D models and animations of proteins and cells for interactive learning.
- Define product vision and execute the business strategy as a founding member.

Product Owner | Systems Engineer — Epsilon Systems Solutions, Largo, FL · May 2020 – October 2021

- Led internal R&D to create new solutions the firm could leverage to win work, including a biology-inspired drone concept for a competitive bid.
- Applied model-based systems engineering to design solutions and implement changes to the codebase.
- Designed system and unit tests to ensure quality of team deliverables.
- Ran Agile/Scrum delivery: backlog grooming, test-driven development planning and stakeholder communication across changing timelines, budgets and requirements.

Graduate Teaching Assistant — University of South Florida, Tampa, FL · January 2014 – May 2020

- Lab TA: delivered weekly lessons; wrote, administered and graded quizzes, exams and lab reports; supervised experiments and enforced lab safety policy.
- Prep TA: managed lab inventory and orders; prepared reagents and equipment; supported other TAs during equipment failures.
- Lecture TA: co-created and graded assignments, tutored students in office hours and assisted with exam administration.

Publications

Peer-reviewed articles

1. Gorman BR, Francis M, Nealon CL, Halladay CW, **Duro N**, Markianos K, Genovese G, Hysi PG, Choquet H, Afshari NA, Li YJ, VA Million Veteran Program, Gaziano JM, Hung AM, Wu WC, Greenberg PB, Pyarajan S, Lass JH, Peachey NS, Iyengar SK. A multi-ancestry GWAS of Fuchs corneal dystrophy highlights the contributions of laminins, collagen, and endothelial cell regulation. *Communications Biology*. 2024;7:418. doi:10.1038/s42003-024-06046-3 (preprint: Research Square, 2023)
2. Kriss CL, **Duro N**, Nadeau OW, Guergues J, Chavez-Chiang O, Culver-Cochran AE, Chaput D, Varma S, Stevens SM Jr. Site-specific identification and validation of hepatic histone nitration in vivo:

implications for alcohol-induced liver injury. *Journal of Mass Spectrometry*. 2021;56(6):e4713. doi:10.1002/jms.4713

3. Delgado JM, **Duro N**, Rogers DM, Tkatchenko A, Pandit SA, Varma S. Molecular basis for higher affinity of SARS-CoV-2 spike RBD for human ACE2 receptor. *Proteins: Structure, Function, and Bioinformatics*. 2021;89(9):1134–1144. doi:10.1002/prot.26086 (preprint: bioRxiv, September 2020)

4. **Duro N**, Varma S. Role of structural fluctuations in allosteric stimulation of paramyxovirus hemagglutinin-neuraminidase. *Structure*. 2019;27(10):1601–1611.e2. doi:10.1016/j.str.2019.07.005

5. **Duro N**, Gjika M, Siddiqui A, Scott HL, Varma S. POPC bilayers supported on nanoporous substrates: specific effects of silica-type surface hydroxylation and charge density. *Langmuir*. 2016;32(26):6766–6774. doi:10.1021/acs.langmuir.6b01155

Conference abstracts

- Said I, Shi Y, Mittal A, Li W, Zheng L, **Duro N**, Devineni P, Khazanov N, Gollub J, Pyarajan S. P304: Using short-read alignments to a pan-genome graph to call variants in challenging medically relevant genes in the Million Veteran Program. ACMG Annual Clinical Genetics Meeting; *Genetics in Medicine Open*. 2026;4:103798. doi:10.1016/j.gimo.2026.103798
- **Duro N**, Varma S. Identifying time-resolved allosteric signaling pathways in proteins using supervised machine learning. Biophysical Society Annual Meeting; *Biophysical Journal*. 2020.
- **Duro N**, Varma S. Structural fluctuations are key to allosteric stimulation of NDV hemagglutinin-neuraminidase. Biophysical Society Annual Meeting; *Biophysical Journal*. 2019.
- **Duro N**, Varma S. Dynamic allostery in regulated entry of Newcastle disease virus into host cells. Biophysical Society Annual Meeting; *Biophysical Journal*. 2017.
- **Duro N**, Dutta P, Varma S. Sialic acid induced conformational ensemble shifts in the hemagglutinin-neuraminidase protein of the Newcastle disease virus. Biophysical Society Annual Meeting; *Biophysical Journal*. 2016.
- **Duro N**, Gjika M, Scott HL, Varma S. Structural and dynamical properties of POPC bilayers supported on nanoporous substrates. Biophysical Society Annual Meeting; *Biophysical Journal*. 2015.

Dissertation

- **Duro N**. Allosteric Control of Proteins: New Methods and Mechanisms. Ph.D. dissertation, University of South Florida, 2020. ProQuest.

Manuscripts in preparation (as listed on the 2021 resume; confirm current status before sending)

- **Duro N**, Varma S. Unscrambling time-resolved allosteric signaling pathways in proteins using machine learning.
- **Duro N***, Drew-Beare J*, Cull D, Varma S. Paramyxovirus attachment proteins: three subfamilies, three different ligand binding responses. (*equal contribution)

Technical Skills

Area	Skills and tools
Genomics and bioinformatics	PacBio HiFi long-read sequencing; de novo genome assembly (hifiasm) and assembly QC; pangenome construction (Minigraph-Cactus, HPRC methods); reference panel development; structural variant discovery; variant filtering and imputation; AGC genome collection archiving
Multi-omics and statistics	Metabolomics and PheWAS analysis; data normalization, imputation and integration; R-based analysis pipelines
Computational biophysics	All-atom molecular dynamics; molecular and homology modeling; statistical thermodynamics; protein–ligand, protein–protein and protein–lipid interactions; supervised machine learning (SVM), graph theory and affinity propagation for allosteric pathway analysis
Programming and workflows	Python, R, C/C++, Bash/Linux; WDL and Snakemake pipelines; Git
High-performance computing	LSF job scheduling, queue and fairshare management; reproducible pipeline engineering on shared clusters
Project and product management	Agile/Scrum; requirements gathering; stakeholder engagement; risk assessment; model-based systems engineering; team leadership and mentoring

Presentations, Certifications and Leadership

Invited and contributed talks

- “Role of Structural Fluctuations in Allosteric Stimulation of Paramyxovirus Hemagglutinin-Neuraminidase.” 63rd Annual Meeting of the Biophysical Society, 2019.
- “Role of Dynamics in the Function of the NDV Attachment Protein.” 37th Annual Meeting of the American Society for Virology, 2018.
- “Role of Dynamics in the Function of the NDV Attachment Protein.” Computational Science Club Student Research Showcase, University of South Florida, 2018.
- “Allosteric Stimulation of the NDV Host Binding Protein.” CMMB Departmental Seminar, University of South Florida, 2017.
- Poster presentations at the Biophysical Society Annual Meeting, 2015–2020 (see Conference abstracts above).

Community leadership and enterpreneurship

- Founding member and Chief of Molecular Modeling, Biological Information OS Inc. (2020–present).

- Founder and operator, PakezDrite.com — bilingual Albanian/English job board serving the Tampa Bay area.
- President, Albanian American National Organization (AANO), Tampa Bay Chapter.
- Secretary, National Albanian Registry.